Jaegun Lee
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Republic of Korea

Jaegun Lee

Research Interests

- Geometric Algorithms
  - Partitioning, Covering

Education

Pohang University of Science and Technology **2022 - present**
M.S./Ph.D.(Integrated) in *Computer Science and Engineering*
Advisor: Hee-Kap Ahn
GPA: 4.04/4.3

Pohang University of Science and Technology **2017 - 2022**
B.S. in *Creative IT Engineering* & *Mathematis*
GPA: 3.81/4.3 (Cum Laude)

Publications

Journal

1. Hwi Kim, **Jaegun Lee**, Hee-Kap Ahn. 2024
   Uniformly monotone partitioning of polygons.
   Theoretical Computer Science.
2. Hwi Kim, **Jaegun Lee**, Hee-Kap Ahn. 2023
   Rectangular Partitions of a Rectilinear Polygon.
   Computational Geometry.

Conference

1. **Jaegun Lee**, Chaeyoon Chung, Hee-Kap Ahn. 2026
   Bichromatic Classifications using Strips.
   The 20th Scandinavian Symposium on Algorithm Theory (SWAT 2026).
2. **Jaegun Lee**, Seokyun Kang, Hyeonseok Lee, Hyeyun Yang, Taehoon Ahn. 2026
   CG#Hunters Approach to Central Triangulation Under Parallel Flip Operations.
   The 42nd International Symposium on Computational Geometry (SoCG 2026).
3. Chaeyoon Chung, **Jaegun Lee**, Hee-Kap Ahn. 2025
   Covering Weighted Points using Unit Squares.
   The 36th International Symposium on Algorithms and Computation (ISAAC 2025).
4. **Jaegun Lee**, Hyojeong An, Hwi Kim, Hee-Kap Ahn. 2025
   Monotone Partitions of Simple Polygons.
   The 36th International Workshop on Combinatorial Algorithms (IWOCA 2025).
5. Taehoon Ahn, **Jaegun Lee**, Byeonguk Kang, Hwi Kim. 2025
   Incremental algorithm and local search for minimum non-obtuse triangulation.
   The 41st International Symposium on Computational Geometry (SoCG 2025).
6. Jongmin Choi, **Jaegun Lee**, Hee-Kap Ahn. 2023
   Efficient k-center algorithms for planar points in convex position.
   The 18th Algorithms and Data Structures Symposium (WADS 2023).
7. Hwi Kim, **Jaegun Lee**, Hee-Kap Ahn. 2022
   Uniformly Monotone Partitioning of Polygons Revisited.
   The 34th Canadian Conference on Computational Geometry (CCCG 2022).

Presentations and talks at workshops

1. Bichromatic Classifications using Strips. 2026
   The 17th Annual Meeting of Asian Association for Algorithms and Computation (AAAC 2026).
2. Monotone Partitions of Simple Polygons. 2025
   The 16th Annual Meeting of Asian Association for Algorithms and Computation (AAAC 2025).
3. Efficient k-center algorithms for planar points in convex position. 2023
   The 23rd Japan–Korea Joint Workshop on Algorithms and Computation (WAAC 2023).

Scholarships & Awards

- Third Place at CG:SHOP SoCG Competition. 2026
- Second Place & First Place in Junior at CG:SHOP SoCG Competition. 2025
- Bang Sung-Yang Graduate Fellowship 2022

Projects

- Auto Nesting
  By Samsung Heavy Industries 2025-2026
- Slab Yard Optimization
  By POSCO DX, Team Leader 2023-2024
- Software Star Lab (Optimal Data Structure and Algorithmic Applications in Dynamic Geometric Environment)
  By Software Star Lab 2022-2025

Patent

1. "Slab Transfer System Using Virtual Slab Yard Simulator", Hee Kap AHN, **Jaegun LEE**, Jae Hoon CHUNG, Chae Yoon CHUNG, Minju SONG
   Korean Patent (Application). 10-2024-0201787 2024
2. "Efficient Optimal Facility Location Determination Methods for Convex Position Demand Points", Hee Kap AHN, Tae Kang EOM, Jong Min CHOI, **Jae Gun LEE**
   Korean Patent. 10-2704974 2024
3. "Efficient Optimal Facility Location Determination Methods for Convex Position Demand Points", Hee Kap AHN, Tae Kang EOM, Jong Min CHOI, **Jae Gun LEE**
   US Patent (Application). 18/496700 2023
4. "METHOD AND APPARATUS FOR CLASSIFYING IMAGE", Chi Won SUNG, **Jae Gun LEE**
   US Patent. 11657594 2023
5. "METHOD AND APPARATUS FOR CLASSIFYING IMAGE", Chi Won SUNG, **Jae Gun LEE**
   Korean Patent. 1021590520000 2020

Academic Activities

Conference Organization

The 26th Korea–Japan Joint Workshop on Algorithms and Computation (WAAC 2026) 2026
The 25th Korean Workshop on Computational Geometry (KWCG 2026) 2025

Official Reviewer of Journals

Computational Geometry: Theory and Applications (CGTA) 2024

Official Reviewer of Conferences

Symposium on Theoretical Aspects of Computer Science (STACS) 2025
International Conference and Workshop on Algorithms and Computation (WALCOM) 2025
International Symposium on Computational Geometry (SoCG) 2024
Theory and Applications of Models of Computation (TAMC) 2024
International Symposium on Algorithms and Computation (ISAAC) 2023

Educational Activities

Teaching Assistant of AI Education Programs for Business

POSCO AI Expert 2022
Youth AI·Bigdata Academy 2022.6, 2022.12, 2023, 2024.2, 2024.8, 2025.4

HuStar Innovation Academy[1] 2022

Teaching Assistant in *Pohang University of Science and Technology*[2]
CSED331 Algorithms 2022, 2023

Teaching Assistant in *Pohang Jecheol High School*
Advanced Thinking Program 2022

Extra Activities

Lab Activities

- KWCG 2025 Local Organizer 2025
- Lab Manager 2025 - 2026

Work Experience

- Polaris3d 2019 - 2020
  - Developer (Out-of-stock Detection Model, demo at CES2020)

Student Organization Activities

- Student Council of department 2018

Club Activities

- Adlib (drama club) 2017 - 2022
  president 2019
- Tachyons (baseball club) 2017 - present

[1] Education program of local governments (Deagu and Gyeongsangbuk-do).
[2] Education program for students of POSTECH.